PCEOT402
ELECTRICAL MACHINES

Module 2

Transformers



Single Phase Transformer

- Practical transformer
- No-load and on-load operation
- Phasor diagrams
- Equivalent circuit
- Voltage Regulation
- Losses
- Condition for maximum efficiency

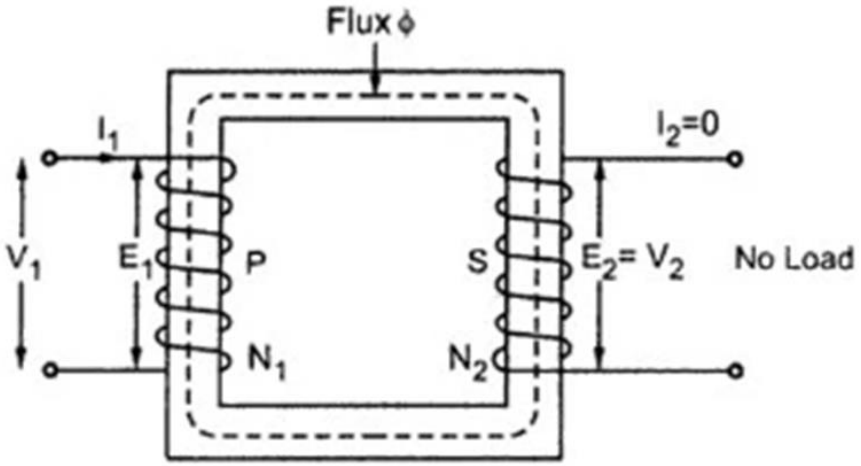
Practical Transformer on No-load

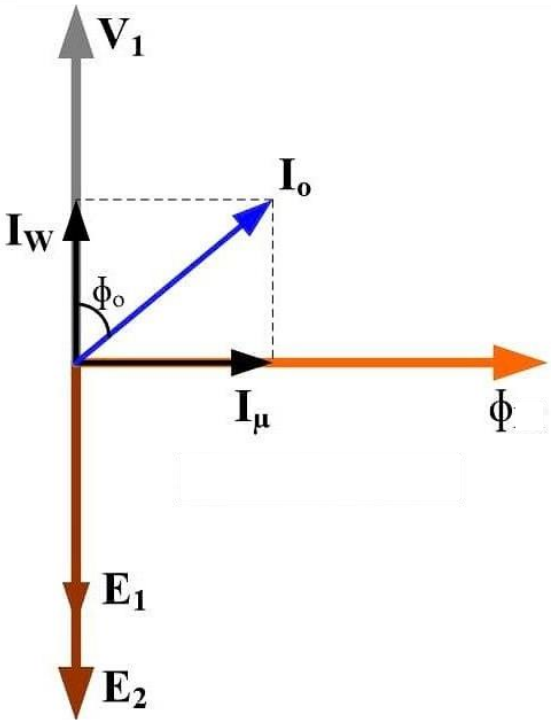
- No-load – secondary circuit is open, i.e., $I_2 = 0$
- actual transformer - iron loss in the core and copper loss in both primary and secondary windings
- Primary current has to supply

  – supply iron loss in the core

  - small copper loss in the primary [$I_2 = 0$; so no copper loss in the secondary winding]
- No-load primary current I_0 lags V_1 by an angle Φ_0, $\Phi_0 < 90^0$

Practical Transformer on No-load





Practical Transformer on No-load

$No\text{-}load\ input\ power\ , W_0 = V_1 I_0 \cos\phi_0$

$where\ \cos\phi_0 = no-load\ primary\ power\ factor$

$2\ components\ of\ I_0\ :\text{-}$

I_w – *active or working or iron loss component*

- *in phase with* V_1
- *supplies iron loss and primary copper loss*

$= I_0 \cos\phi_0$

I_μ – *magnetising component*

- *in quadrature with* V_1
- *produces alternating flux in the core*

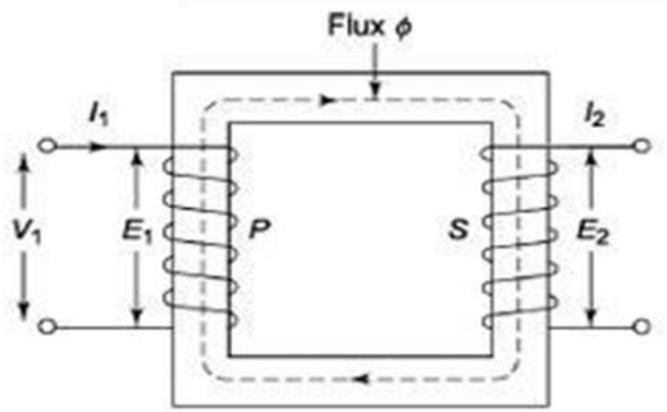
$= I_0 \sin\phi_0$

$I_0 = \sqrt{I_\mu^{\ 2} + I_w^{\ 2}}$

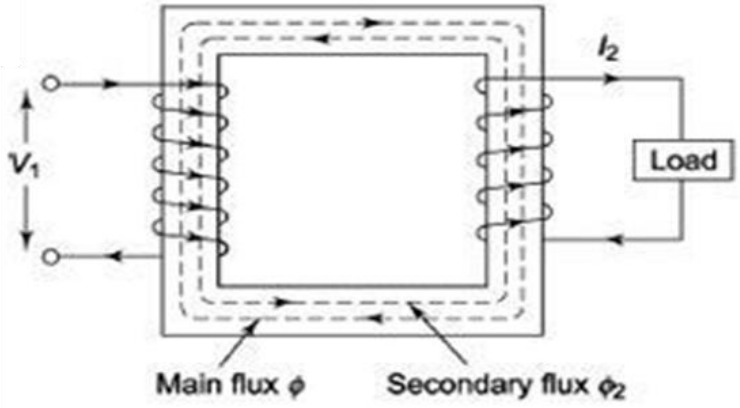
Practical Transformer on No-load

- I_0 is very small , about 1% of the full load current
- Since I_0 is very small , the no-load primary copper loss is very small
- i.e., **no-load primary input power is equal to the iron loss in the transformer**

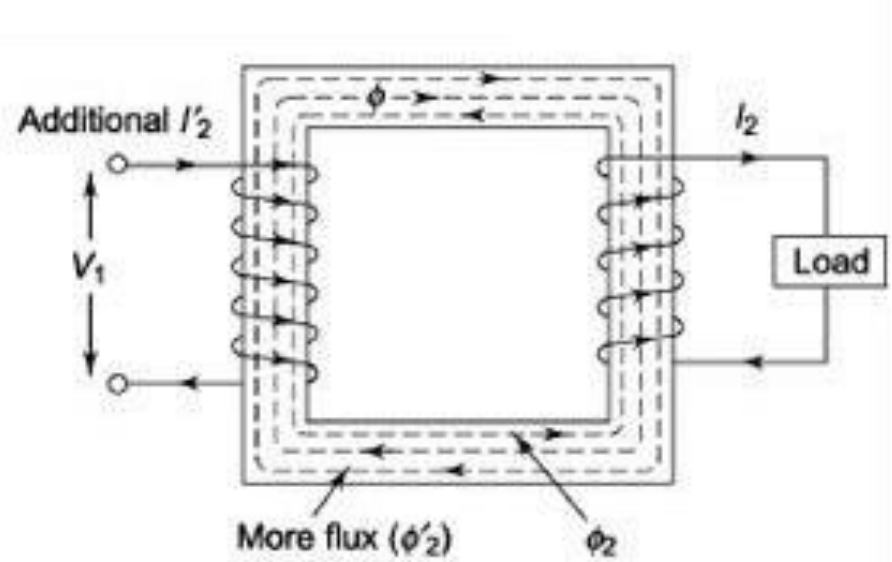
Practical Transformer on Load



(a) No-load condition



(b) ϕ_2 Opposes ϕ



(c) Primary draws more current

Actual Transformer on Load (Flux constant)

Practical Transformer on Load

- When loaded, I_2 is set up
- Magnitude & phase of I_2 wrt V_2 – determined by the characteristics of the load
- I_2 in phase with V_2 – when load is resistive
- I_2 lags V_2 – when load is inductive
- I_2 leads V_2 – when load is capacitive
- I_2 sets up an mmf $N_2 I_2$ –and also its own flux Φ_2
- Φ_2 is in opposition to main flux Φ
- Φ *is due to* I_0

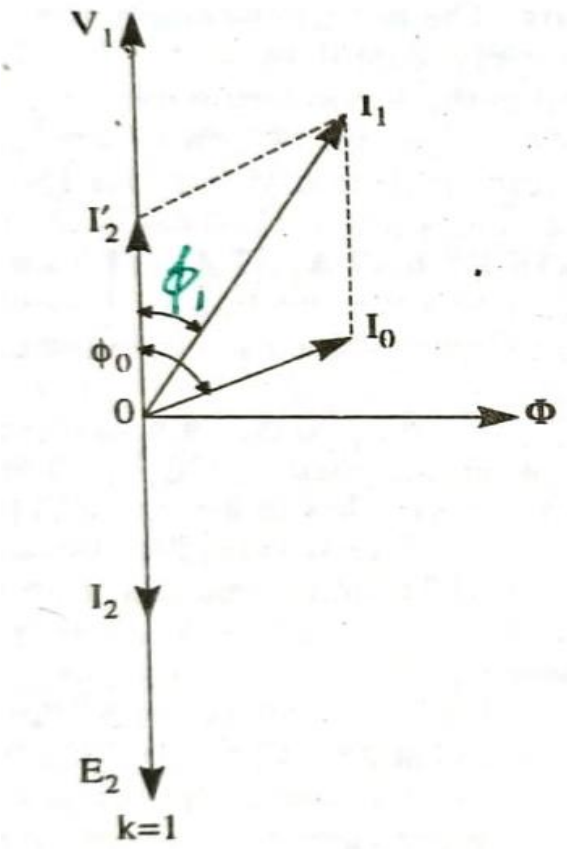
Practical Transformer on Load

- $N_2 I_2$ –called demagnetising ampere turns
- Φ_2 weakens Φ momentarily
- Hence E_1 tends to be reduced.
- V_1 gains upper hand and causes more current to flow in the primary
- Let I_2' be the additional primary current – called load component of primary current
- I_2' is in antiphase with I_2
- I_2' sets up an additional primary mmf $N_1 I_2'$ and its own flux Φ_2'

Practical Transformer on Load

- Φ_2' is in opposition to Φ_2 and in the same direction of main flux Φ
- Φ_2' is equal to Φ_2 and they cancel each other
- Magnetic effects of I_2 is neutralized by additional primary current I_2'
- **Whatever be the load conditions the net flux in the core is same as that at no-load**
- **Therefore core loss is also practically same under all load conditions**

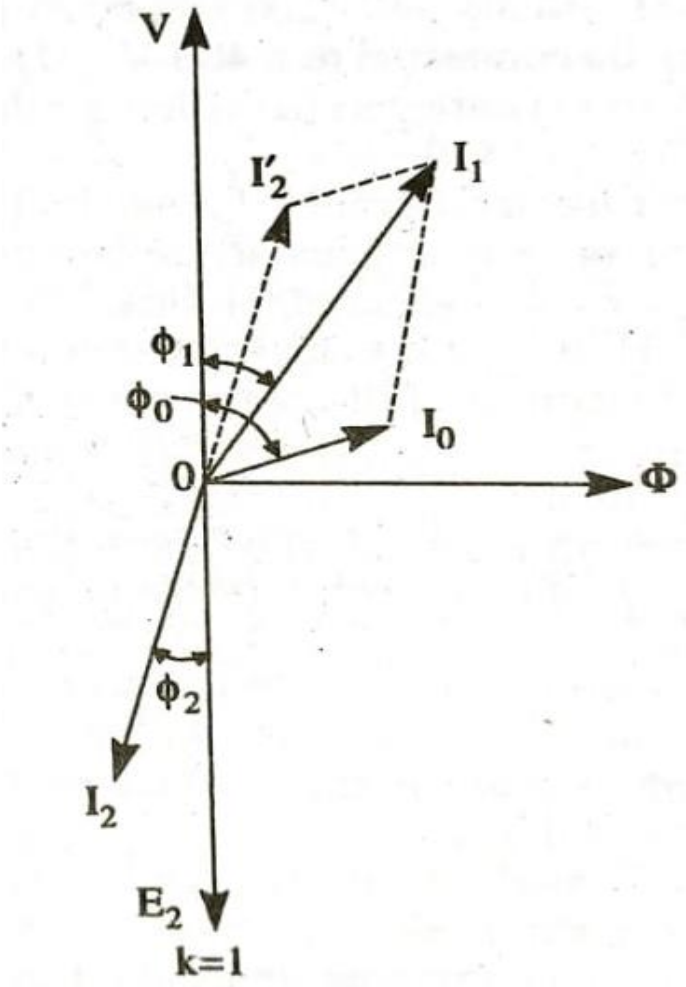
Phasor Diagram of Transformer (with no winding resistance) on Resistive Load-



Transformer (with no winding resistance) on Resistive Load- Phasor Diagram

- Voltage transformation ratio is assumed to be unity
- For resistive load I_2 will be in phase with E_2
- I_2' is antiphase with I_2 and equal in magnitude as K=1
- I_1 –vector sum of I_0 and I_2'
- I_1 lags V_1 by an angle Φ_1

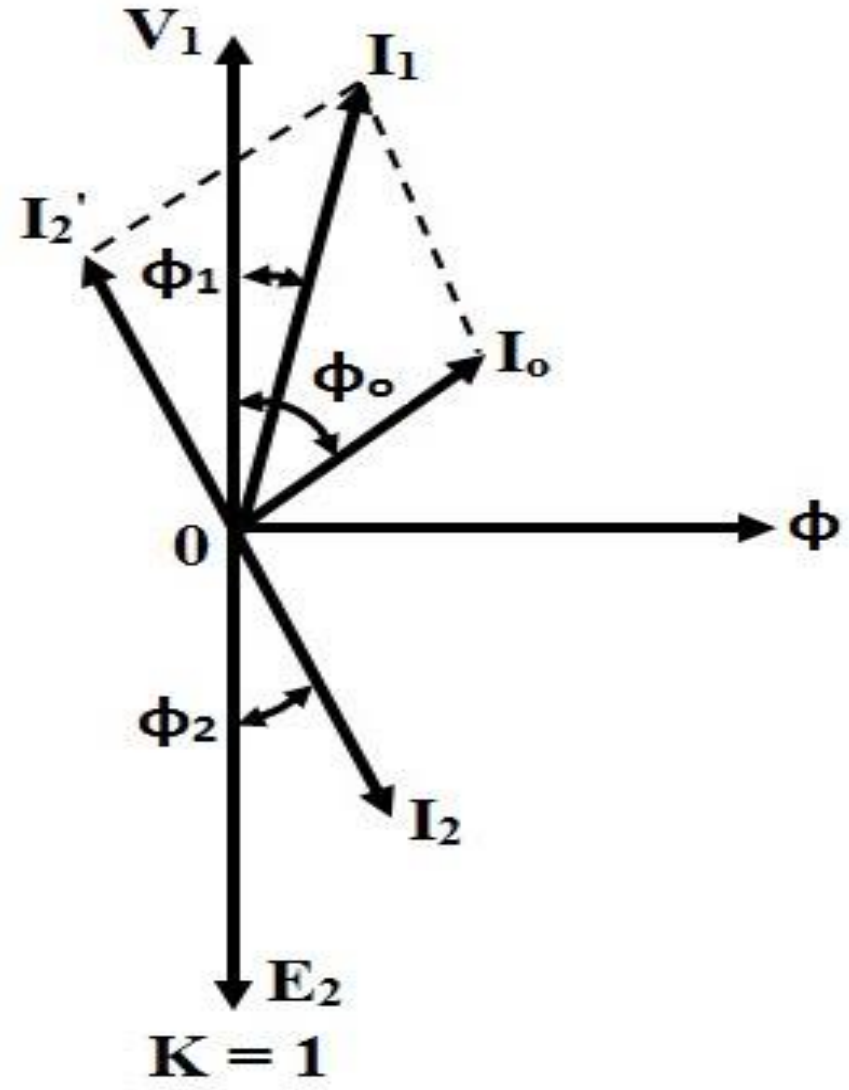
Transformer (with no winding resistance) on Inductive Load- Phasor Diagram



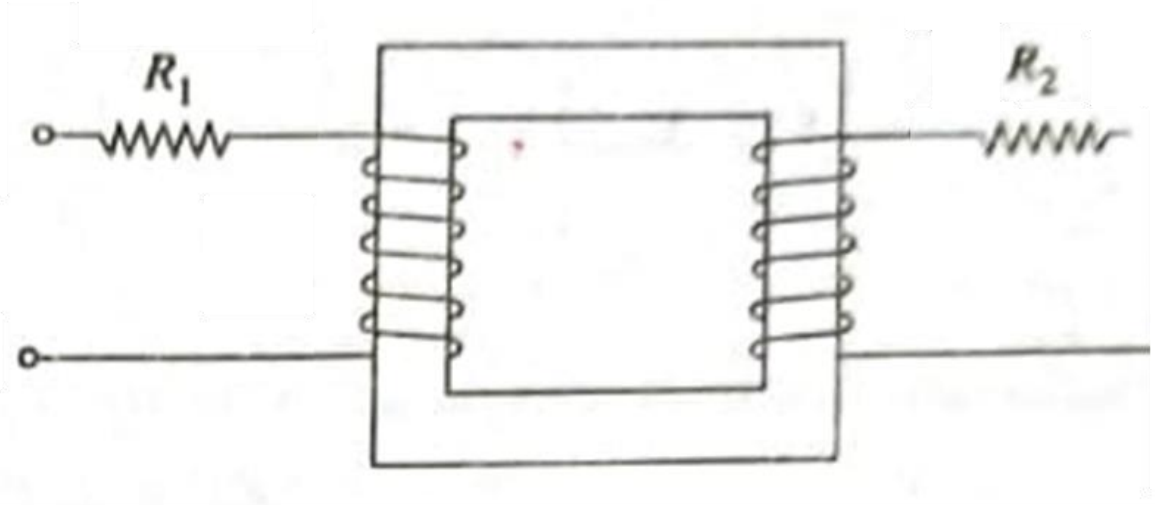
Transformer (with no winding resistance) on Inductive Load

- I_2 lags V_2 by an angle Φ_2
- I_2' is antiphase with I_2 and equal in magnitude as K=1
- I_1 –vector sum of I_0 and I_2'
- I_1 lags V_1 by an angle Φ_1
- Φ_1 is slightly greater than Φ_2.

Transformer (with no winding resistance) on Capacitive Load- Phasor Diagram



Transformer with Resistance & No Magnetic Leakage



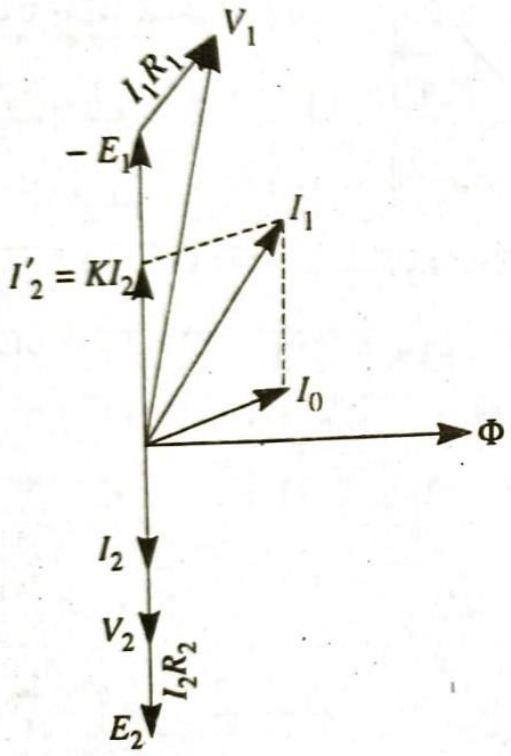
Transformer with Resistance & No Magnetic Leakage

Due to the winding resistance there will be some voltage drop in the 2 windings

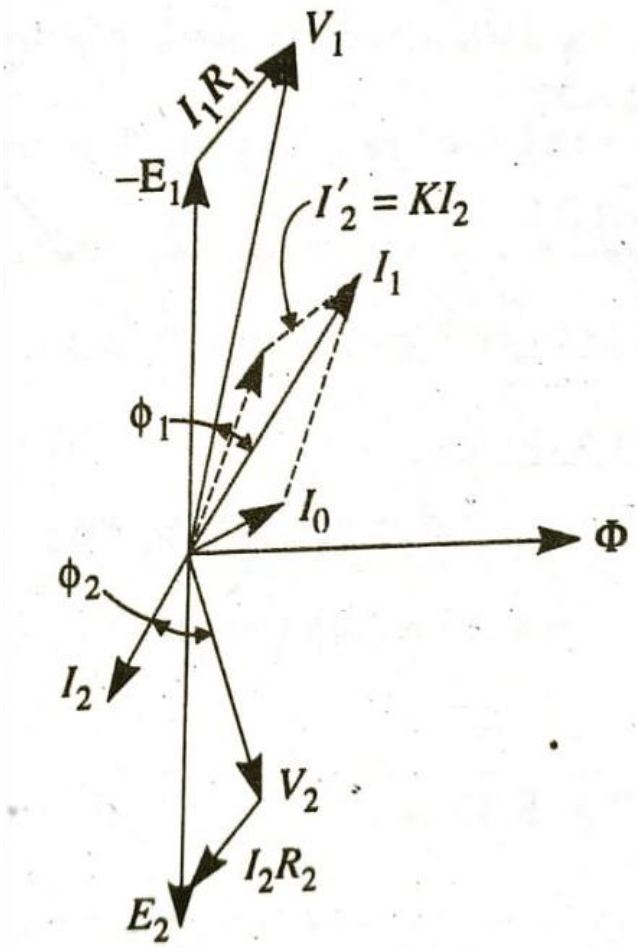
$$E_1 = V_1 - I_1 R_1$$

$$V_2 = E_2 - I_2 R_2$$

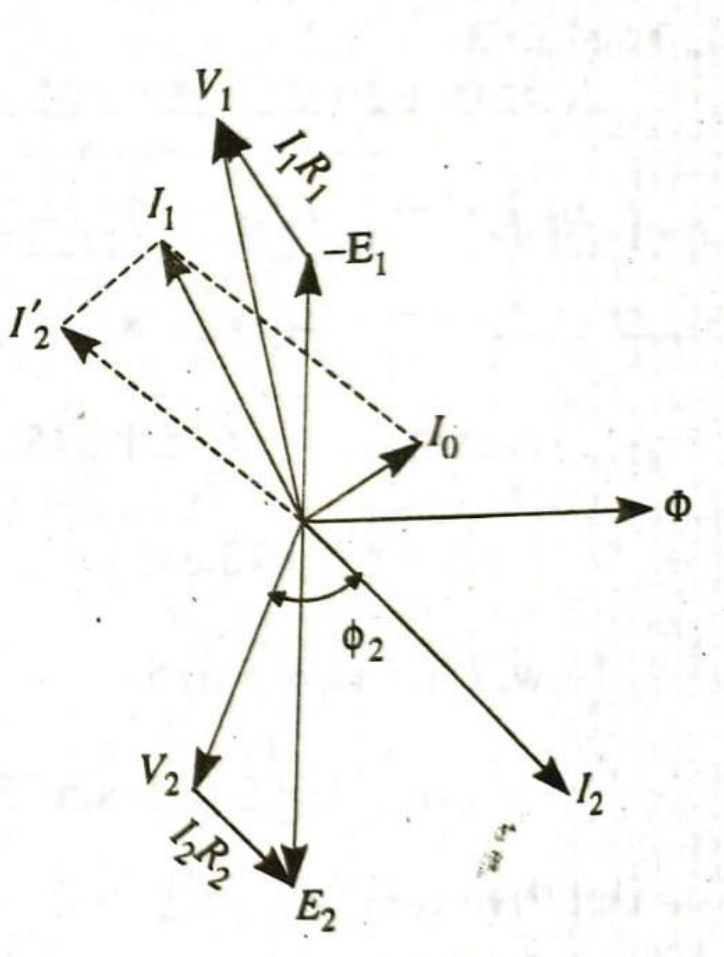
Transformer with Resistance & No Magnetic Leakage- Resistive Load



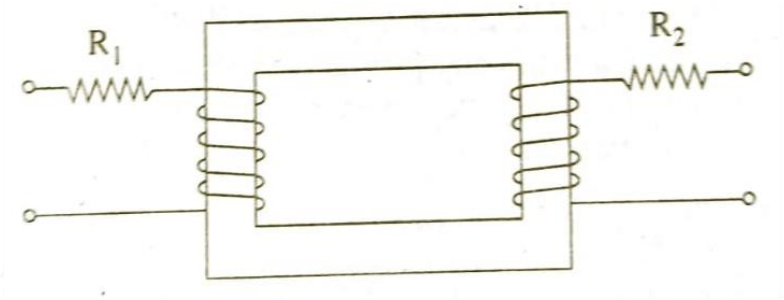
Transformer with Resistance & No Magnetic Leakage- Inductive Load



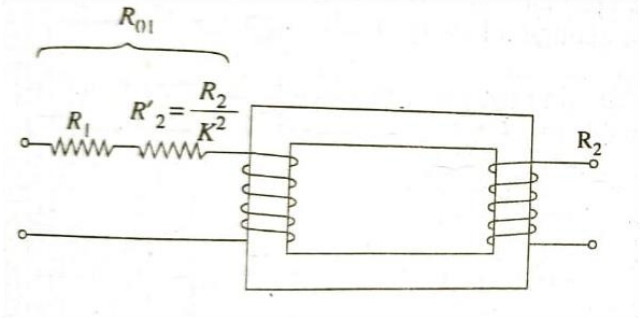
Transformer with Resistance & No Magnetic Leakage-Capacitive Load



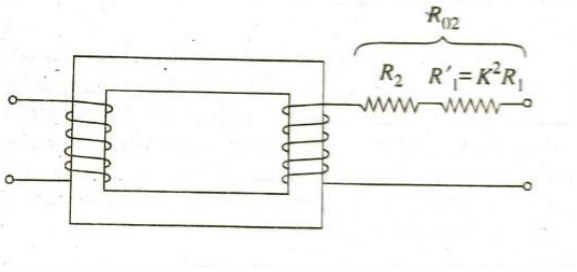
Transformer- Equivalent Resistance



Transformer with primary & secondary resistances shown externally



Transformer with all resistances concentrated in the primary



Transformer with all resistances concentrated in the secondary

Equivalent Resistance

- Resistance of any of the windings can be transferred to any one of the 2 windings
- This makes the calculation simple
- Any resistance transferred to the primary is divided by K^2.
- i.e., R_2 in secondary is equivalent to $R_2' = R_2 / K^2$ in primary, where R_2' is the equivalent secondary resistance referred to primary
- Any resistance transferred to the secondary is multiplied by K^2.
- i.e., R_1 in primary is equivalent to $R_1' = R_1 K^2$ in secondary, where R_1' is the equivalent primary resistance referred to secondary

Equivalent Resistance

Copper loss in secondary winding $= \mathrm{I}_2^2 R_2$

$R_2^{'}$ *is the equivalent resistance in the primary*

which would've caused the same loss as R_2 *in secondary*

$$\mathrm{I}_1^2 R_2^{'} = \mathrm{I}_2^2 R_2 \qquad \text{or } R_2^{'} = \left(\frac{\mathrm{I}_2}{\mathrm{I}_1}\right)^2 R_2 = R_2 / K^2$$

$$\because \frac{I_2}{I_1} = \frac{1}{K}$$

Equivalent Resistance

equivalent primary resistance referred to secondary ,

$$R_1^{'} = K^2 R_1$$

equivalent secondary resistance referred to primary ,

$$R_2^{'} = R_2 / K^2$$

equivalent resistanace of the transformer referred to primary,

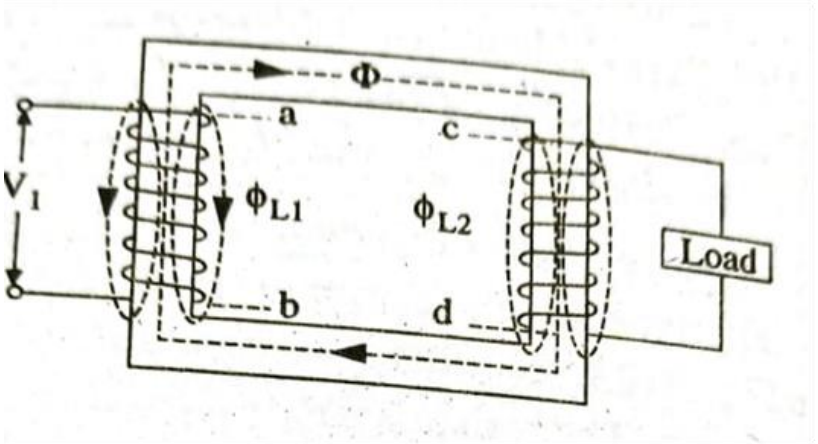
$$R_{01} = R_1 + R_2^{'} = R_1 + R_2 / K^2$$

equivalent resistanace of the transformer referred to secondary,

$$R_{02} = R_2 + R_1^{'} = R_2 + K^2 R_1$$

Magnetic Leakage

- In the case of an ideal transformer all the flux linking with the primary also links the secondary winding
- In actual transformer it is not so
- Φ_{L1}(or Φ_{L2}) completes the magnetic circuit passing through air rather than around the core



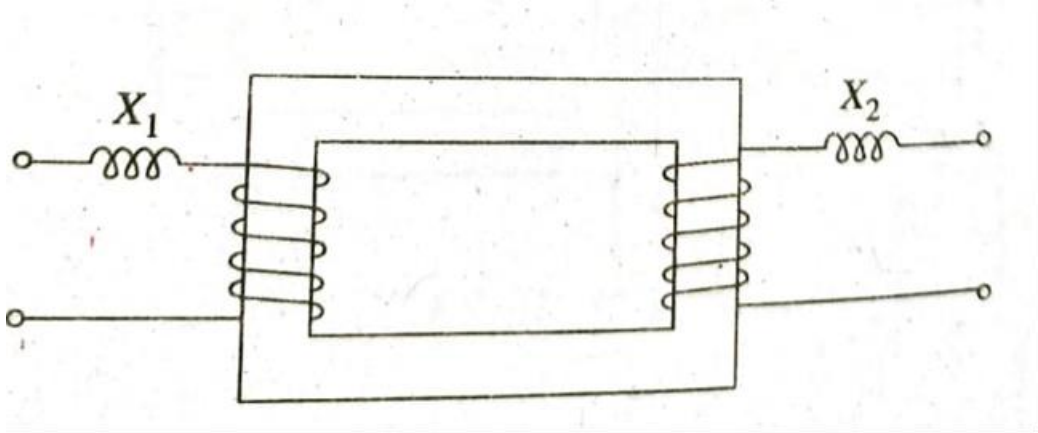
Magnetic Leakage

- Leakage flux is produced when the mmf due to primary ampere turns lying between points ‘a’ and ‘b’ acts along the leakage paths
- Φ_{L1} links with primary winding alone
- Φ_{L1} is in phase with I_1 .
- Φ_{L1} induces an emf e_{L1} in the primary
- secondary ampere turns between points ‘c’ and ‘d’ sets up Φ_{L2} .
- Φ_{L2} links with secondary winding alone
- Φ_{L2} is in phase with I_2 .
- Φ_{L2} induces an emf e_{L2} in the secondary

Magnetic Leakage

- At no load and light loads, leakage flux is small as the ampere turns are small
- When load is increased the currents will set up large mmfs which increase the leakage flux
- The effect of leakage flux can be represented by an inductance in series with the winding
- A transformer with magnetic leakage – ideal transformer with inductive coils in series with the winding

Magnetic Leakage



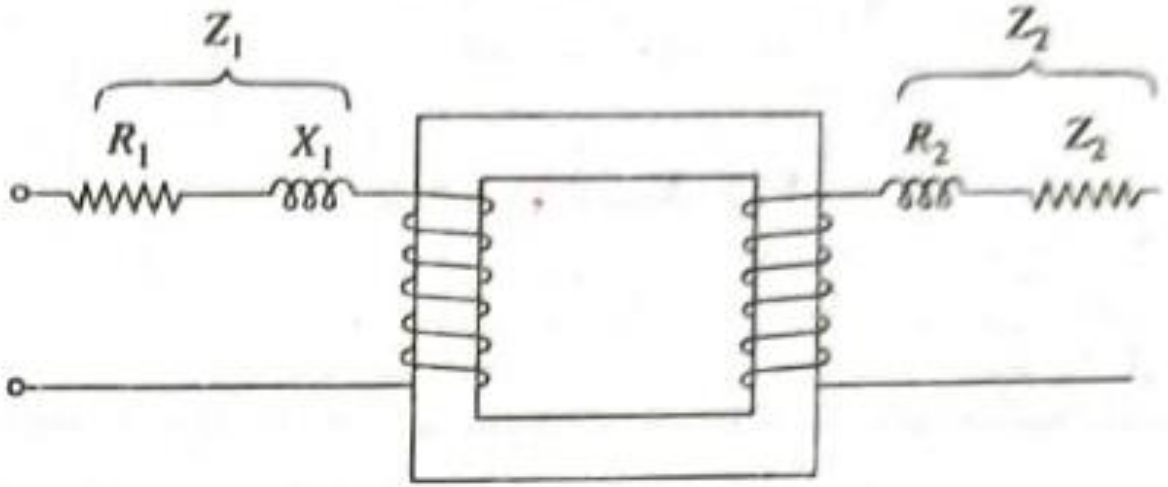
Magnetic Leakage

- Primary leakage reactance, $X_1 = e_{L1}/I_1$
- Secondary leakage reactance, $X_2 = e_{L2}/I_2$

Leakage flux:-

- Links with one winding only
- Does not contribute to the transfer of energy from primary to secondary
- Interleaving of the primary & the secondary windings minimises the leakage flux

Transformer with Resistance & Magnetic Leakage



Transformer with Resistance & Magnetic Leakage

$$\textit{Primary impedance}, Z_1 = \sqrt{(R_1^2 + \mathrm{X}_1^2)}$$

$$\textit{Secondary impedance}, Z_2 = \sqrt{(R_2^2 + \mathrm{X}_2^2)}$$

$$V_1 = E_1 + I_1(R_1 + jX_1) = E_1 + I_1 Z_1$$

$I_1 X_1$ *is usually 1 or 2% of* V_1

$$E_2 = V_2 + I_2(R_2 + jX_2) = E_2 + I_2 Z_2$$

Leakage Reactance

- Leakage reactance of any of the windings can be transferred to any one of the 2 windings
- Any reactance transferred to the primary is divided by K^2.
- i.e., R_2 in secondary is equivalent to $R_2' = R_2 / K^2$ in primary, where R_2' is the equivalent secondary resistance referred to primary
- Any resistance transferred to the secondary is multiplied by K^2.
- i.e., R_1 in primary is equivalent to $R_1' = R_1 K^2$ in secondary, where R_1' is the equivalent primary resistance referred to secondary

Equivalent Leakage Reactance

Leakage reactances can also be transferred from one winding to another.

equivalent primary reactance referred to secondary ,

$$X_1' = K^2 X_1$$

equivalent secondary reactance referred to primary ,

$$X_2' = X_2 / K^2$$

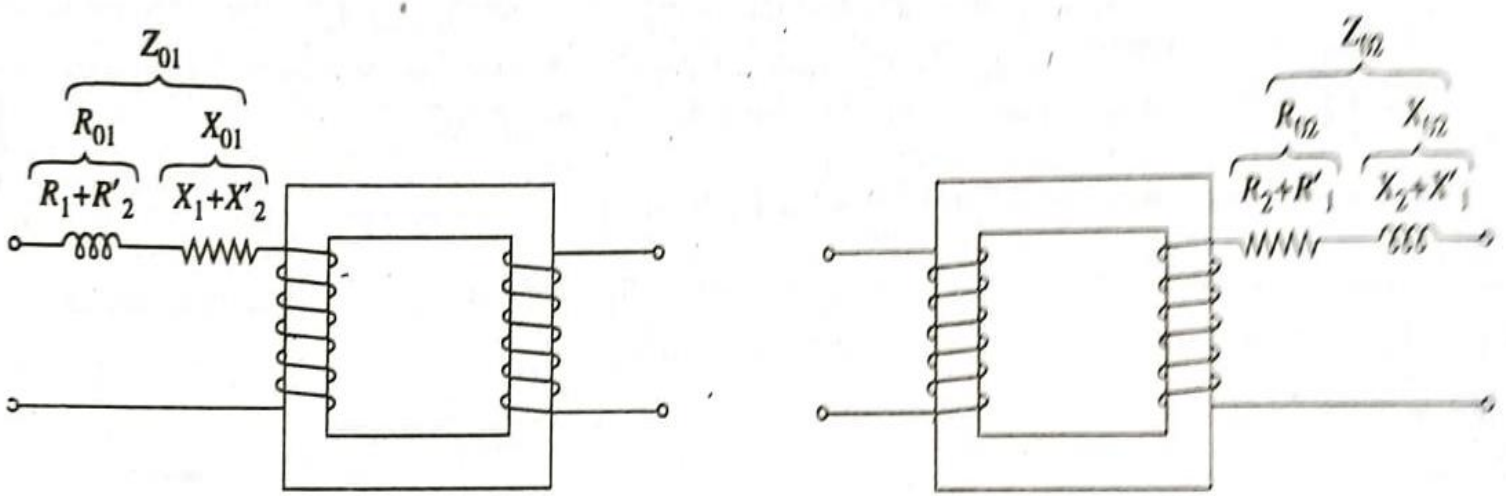
equivalent reactance of the transformer referred to primary,

$$X_{01} = X_1 + X_2' = X_1 + X_2 / K^2$$

equivalent reactance of the transformer referred to secondary,

$$X_{02} = X_2 + X_1' = X_2 + K^2 X_1$$

Transformer with Resistance & Magnetic Leakage- Total Impedance



Transformer with Resistance & Magnetic Leakage- Total Impedance

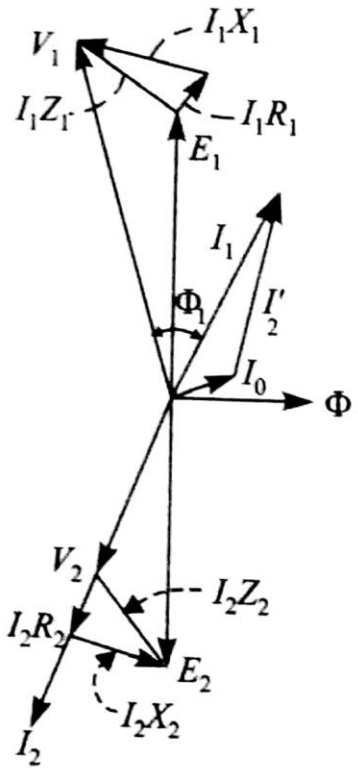
Total impedance of the transformer referred to primary,

$$Z_{01} = \sqrt{(R_{01}^2 + X_{01}^2)}$$

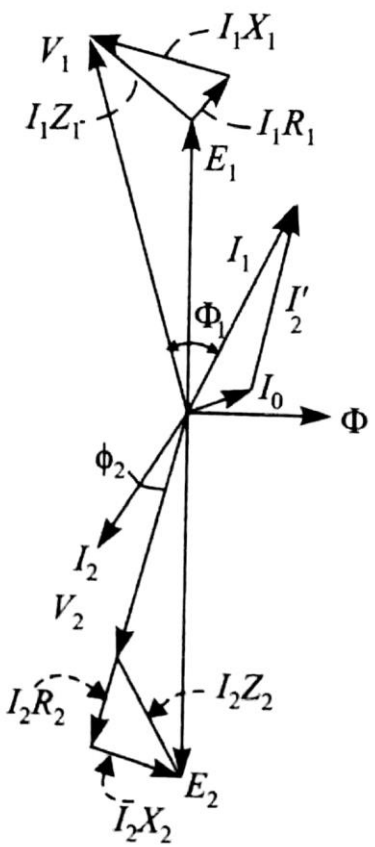
Total impedance of the transformer referred to secondary,

$$Z_{02} = \sqrt{(R_{02}^2 + X_{02}^2)}$$

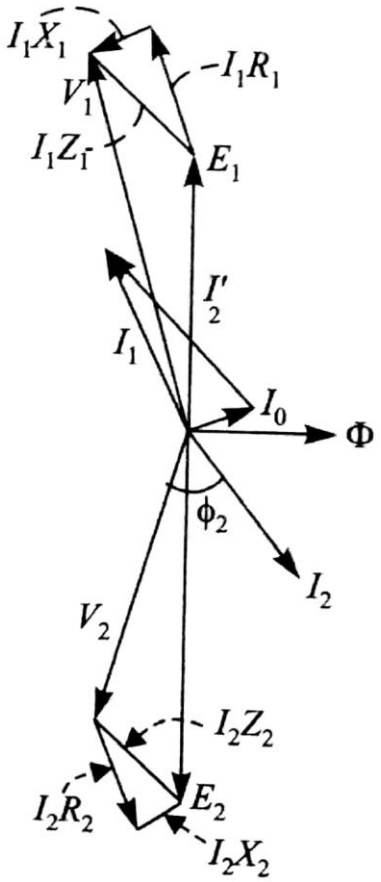
Phasor diagram Transformer with Resistance & Magnetic Leakage



Resistive Load

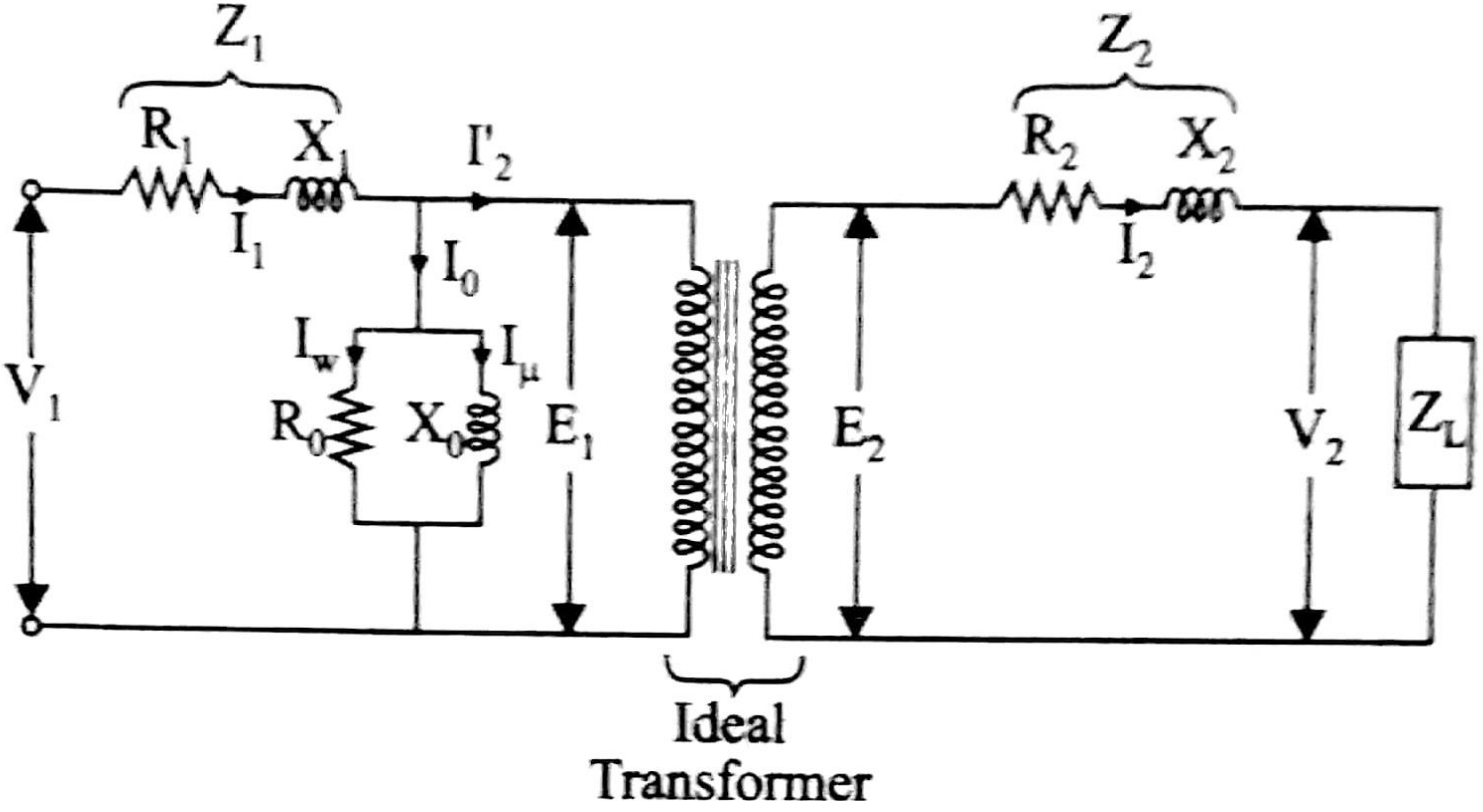


Inductive Load



Capacitive Load

Transformer Equivalent Circuit



Voltage Regulation

- The voltage regulation can be defined in two ways
    - Regulation Down
    - Regulation up.
- Regulation down: This is defined as ” the change in terminal voltage when a load current at any power factor is applied, expressed as a fraction of the no-load terminal voltage”.

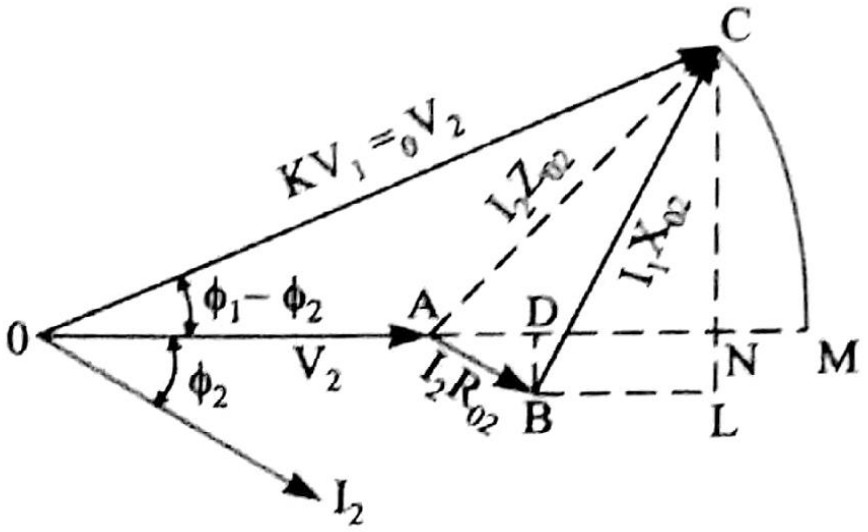
$$Regulation = \frac{|V_{nl}| - |V_l|}{|V_{nl}|}$$

Voltage Regulation

Approximate voltage drop

$$= AN = AD + DN$$

$$= I_2 R_{02} \cos\phi + I_2 X_{02} \sin\phi$$



The approximate voltage drop for *leading* power factor becomes

$$(I_2 R_{02} \cos\phi - I_2 \pm X_{02} \sin\phi)$$

In general, approximate voltage drop is $(I_2 R_{02} \cos\phi \pm I_2 X_{02} \sin\phi)$

It may be noted that approximate voltage drop as referred to primary is

$$(I_1 R_{01} \cos\phi \pm I_1 X_{01} \sin\phi)$$

$$\% \text{ voltage drop in secondary is} = \frac{I_2 R_{02} \cos\phi \pm I_2 X_{02} \sin\phi}{{}_0V_2} \times 100$$

Voltage Regulation

$$Regulation = \frac{|V_{nl}| - |V_l|}{|V_{nl}|}$$

Regulation = $$\frac{I_2 R_{02} \cos \phi \pm I_2 X_{02} \sin \phi}{{}_0V_2} \times 100$$

Losses in a Transformer

- Core loss (iron loss)
    - Hysterisis Loss
    - Eddy Current Loss
- Copper loss

Efficiency of Transformer

$$\text{efficiency} = \frac{\text{output}}{\text{input}}$$

$$\text{efficiency} = \frac{\text{output}}{\text{output + losses}} = \frac{\text{output}}{\text{output + Cu loss + iron loss}}$$

$$\eta = \frac{\text{input} - \text{losses}}{\text{input}} = 1 - \frac{\text{losses}}{\text{input}}$$

Condition for max. efficiency

$$\text{Cu loss} = I_1^2 R_{01} \quad \text{or} \quad I_2^2 R_{02} = W_{cu}$$

$$\text{Iron loss} = \text{hysteresis loss} + \text{eddy current loss} = W_h + W_e = W_i$$

Considering primary side,

$$\text{Primary input} = V_1 I_1 \cos \phi_1$$

$$\eta = \frac{V_1 I_1 \cos \phi_1 - \text{losses}}{V_1 I_1 \cos \phi_1} = \frac{V_1 I_1 \cos \phi_1 - I_1^2 R_{01} - W_i}{V_1 I_1 \cos \phi_1}$$

$$= 1 - \frac{I_i R_{01}}{V_1 \cos \phi_1} - \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

Differentiating both sides with respect to I_1, we get

$$\frac{d\eta}{dI_1} = 0 - \frac{R_{01}}{V_1 \cos \phi_1} + \frac{W_i}{V_1 I_1^2 \cos \phi_1}$$

For η to be maximum, $\frac{d\eta}{dI_1} = 0$. Hence, the above equation becomes

$$\frac{R_{01}}{V_1 \cos \phi_1} = \frac{W_i}{V_1 I_1^2 \cos \phi_1} \quad \text{or} \quad W_i = I_1^2 R_{0r} \quad \text{or} \quad I_2^2 R_{02}$$

or

$$\text{Cu loss} = \text{Iron loss}$$